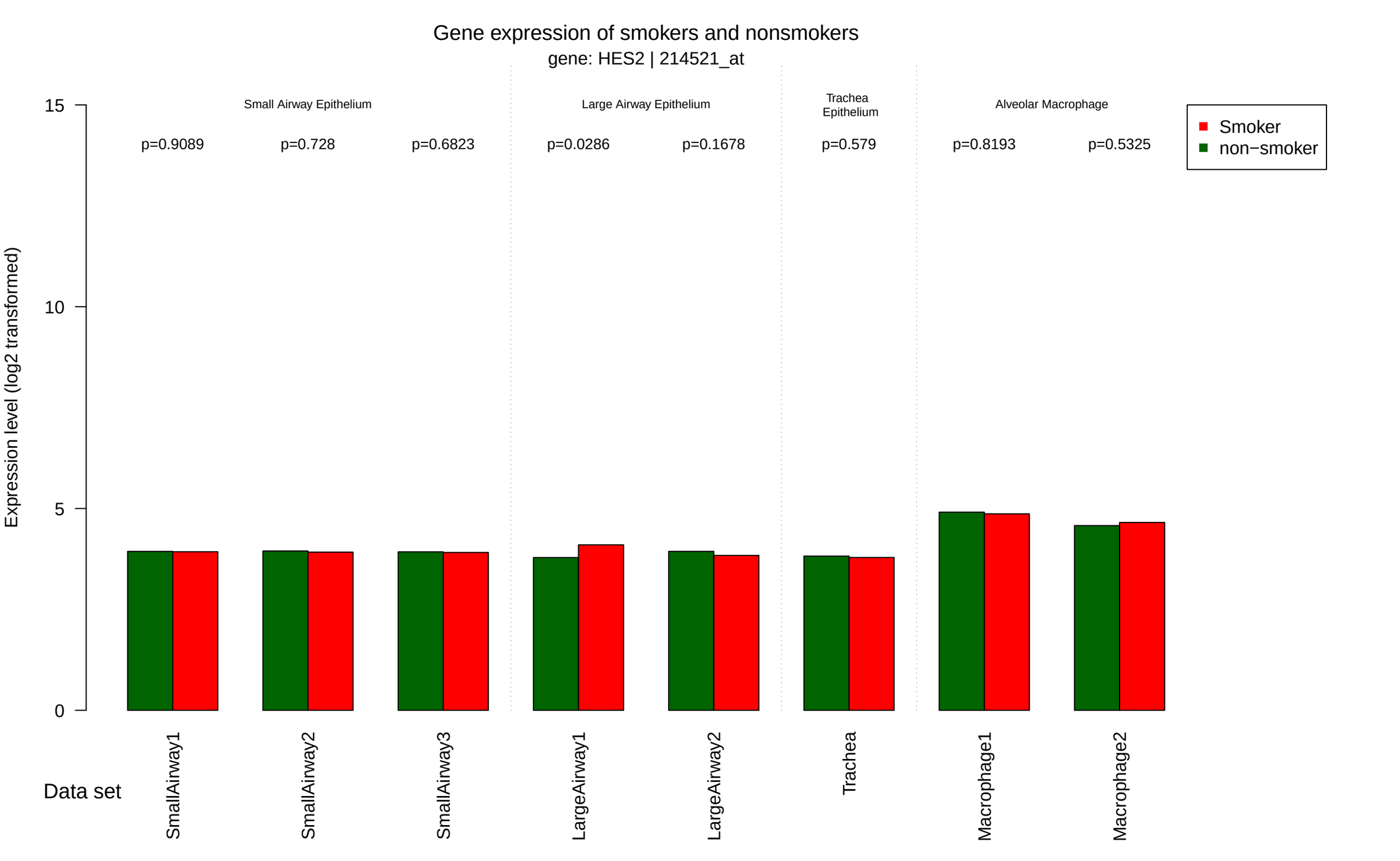
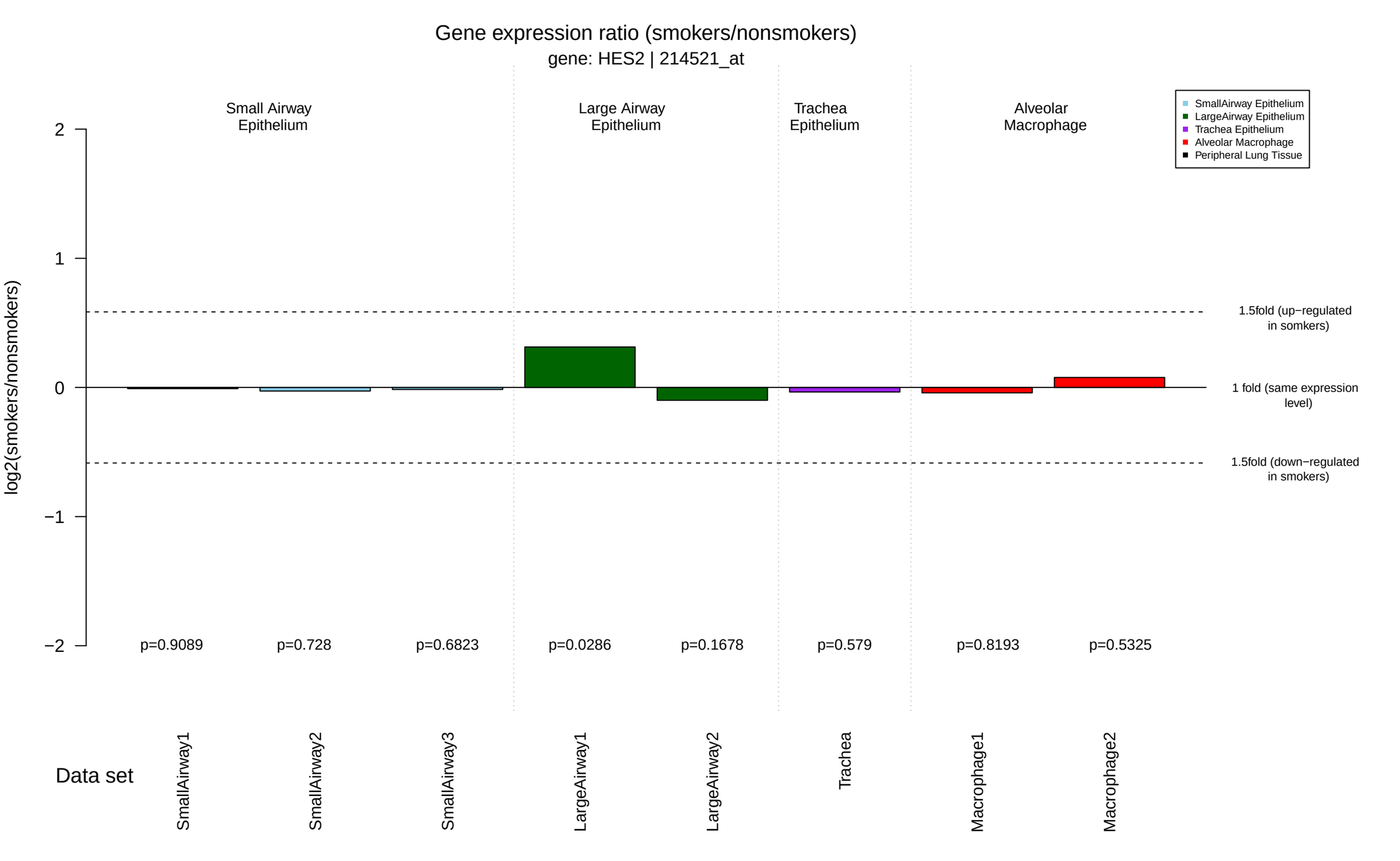
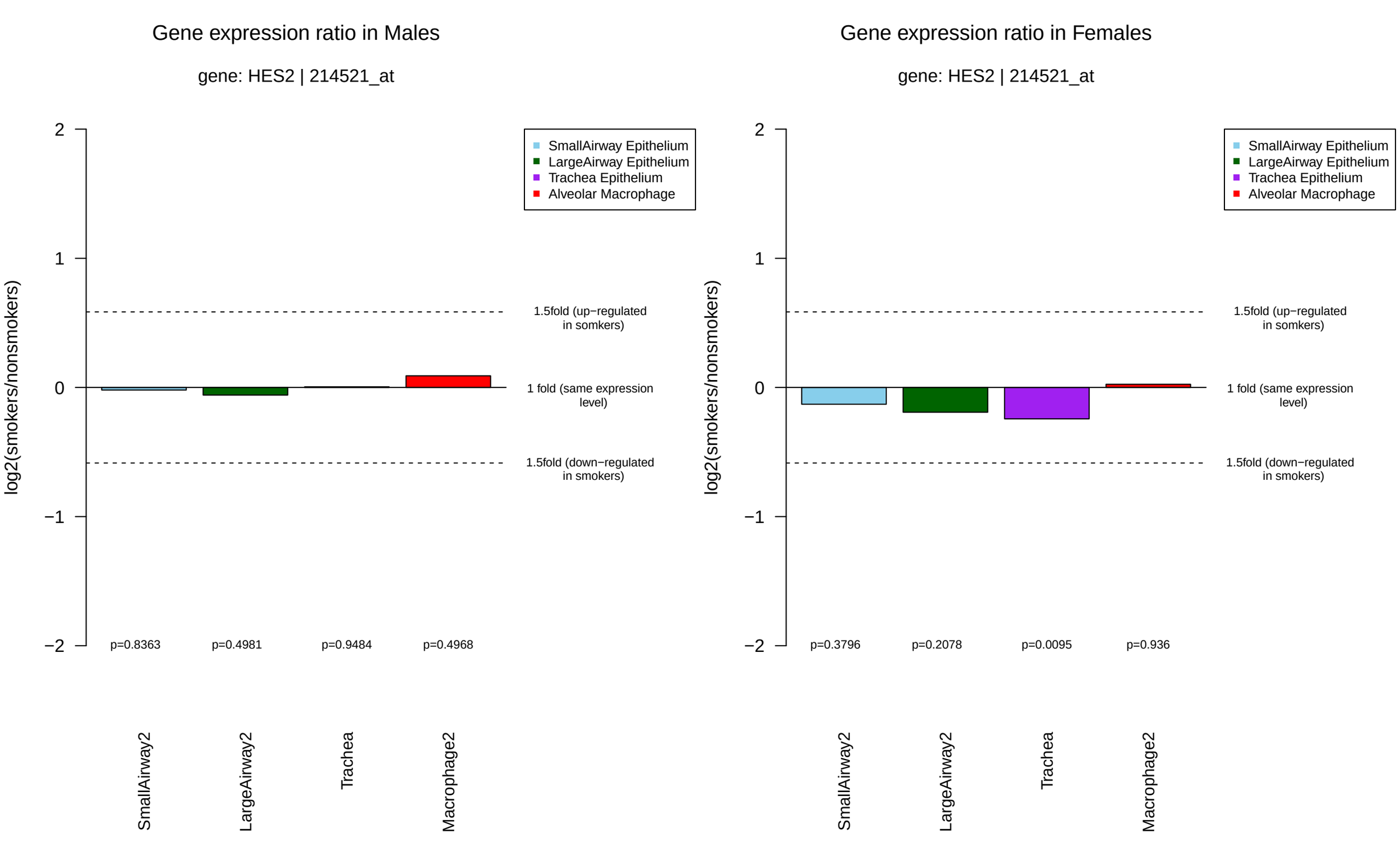
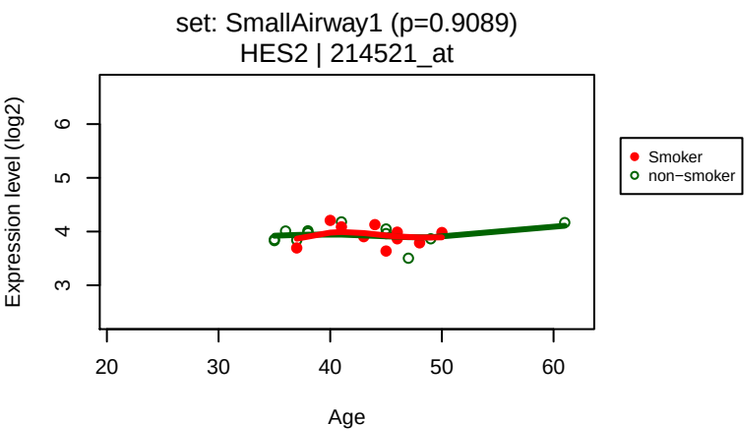
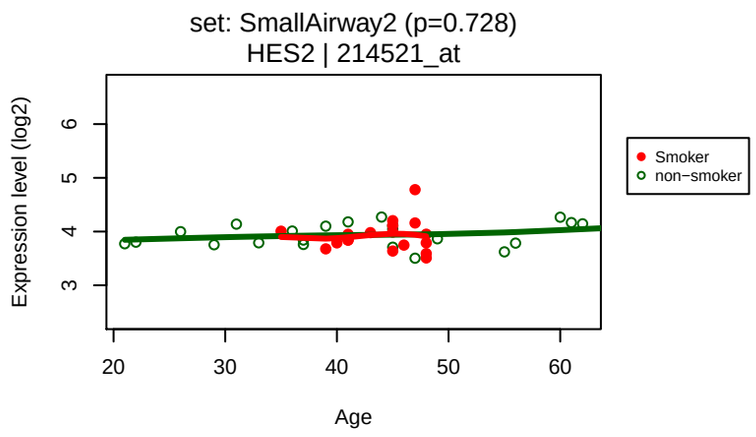
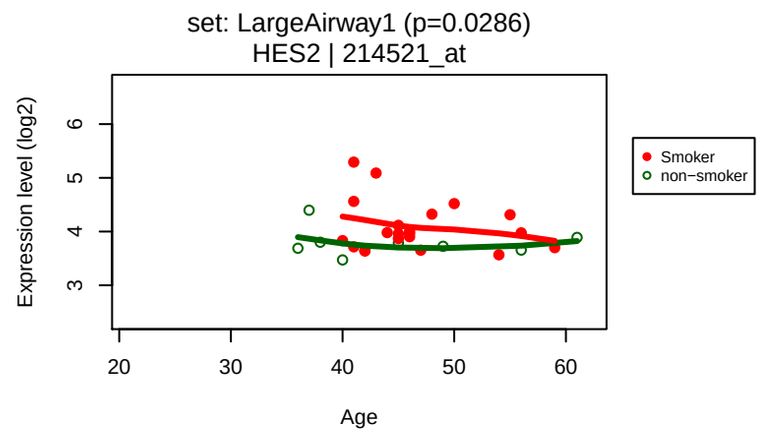

Gene expression of smokers and nonsmokers
gene: HES2 | 214521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9089
p=0.728
p=0.6823
p=0.0286
p=0.1678
p=0.579
p=0.8193
p=0.5325
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

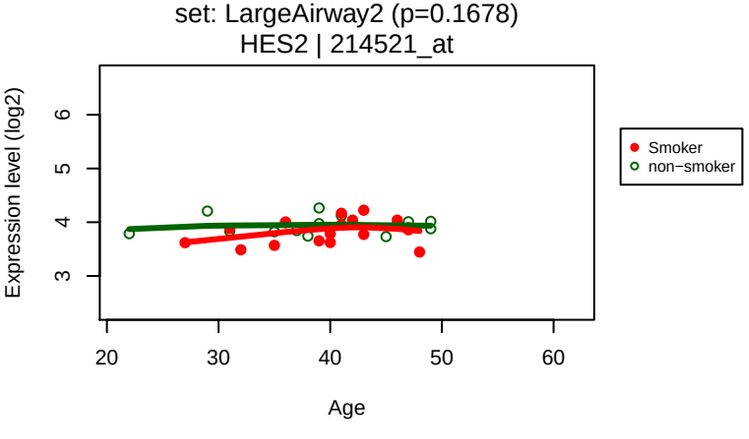

Gene expression ratio (smokers/nonsmokers)
gene: HES2 | 214521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9089
p=0.728
p=0.6823
p=0.0286
p=0.1678
p=0.579
p=0.8193
p=0.5325
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

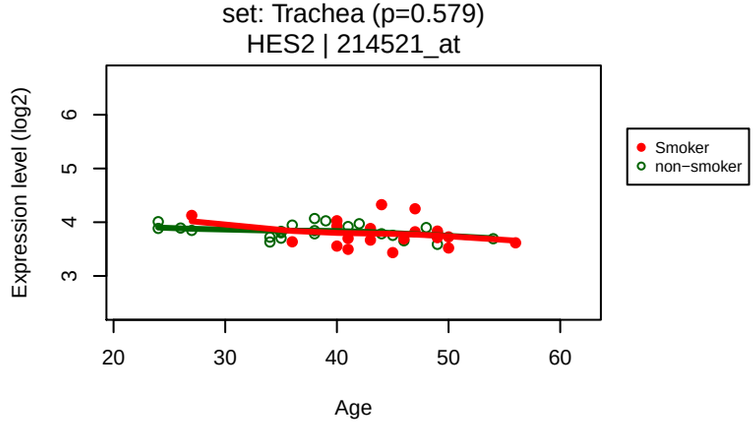

Gene expression ratio in Males
gene: HES2 | 214521_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8363
p=0.4981
p=0.9484
p=0.4968
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HES2 | 214521_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3796
p=0.2078
p=0.0095
p=0.936
SmallAirway2
LargeAirway2
Trachea
Macrophage2

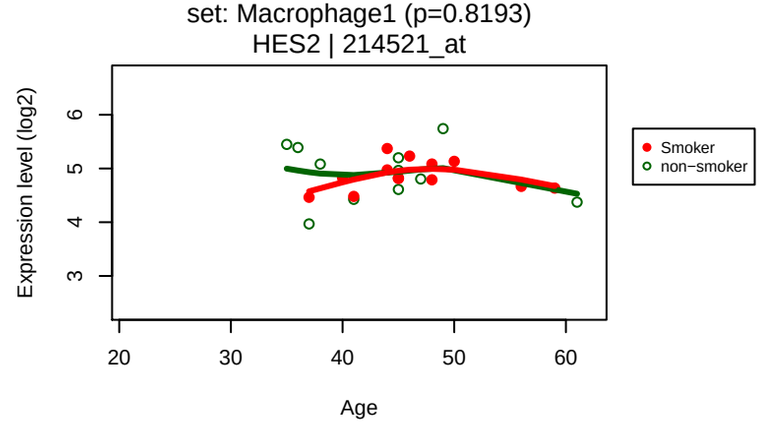

set: SmallAirway1 (p=0.9089)
HES2 | 214521_at
Expression level (log2)
Age
Smoker
non−smoker

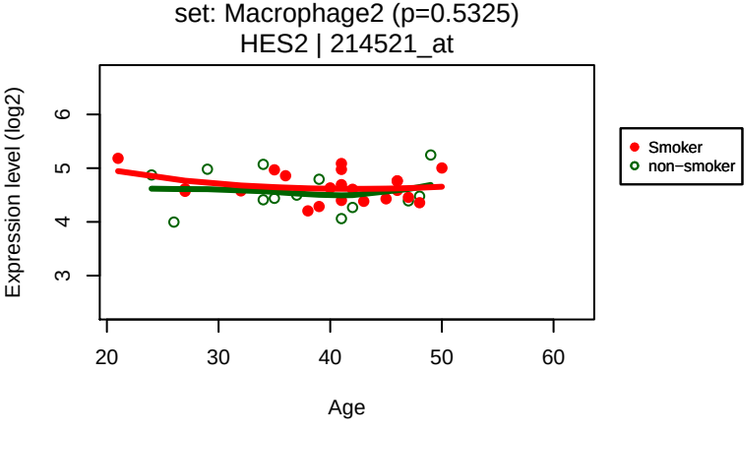

set: SmallAirway2 (p=0.728)
HES2 | 214521_at
Expression level (log2)
Age
Smoker
non−smoker

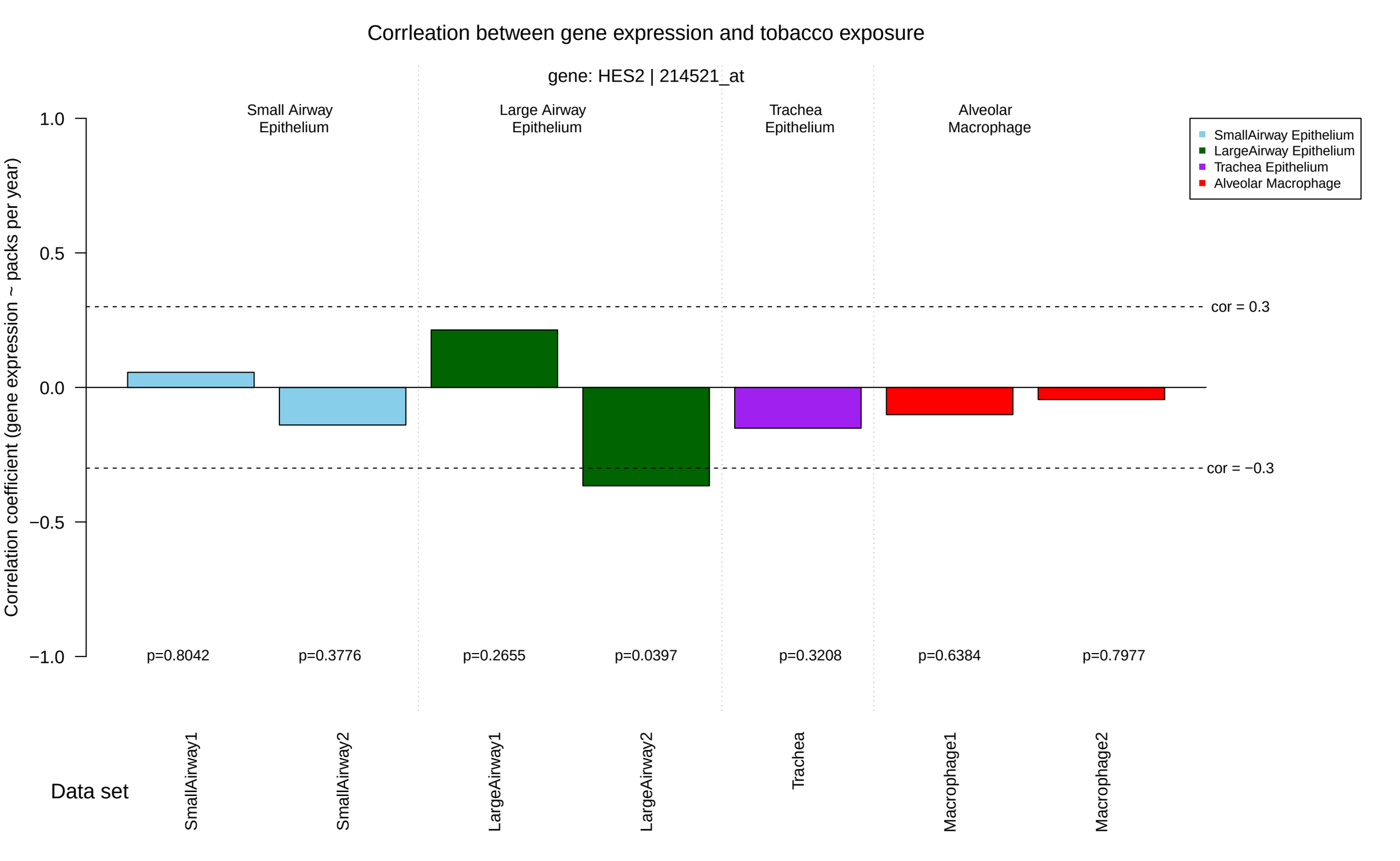

set: LargeAirway1 (p=0.0286)
HES2 | 214521_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1678)
HES2 | 214521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.579)
HES2 | 214521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8193)
HES2 | 214521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5325)
HES2 | 214521_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HES2 | 214521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8042
p=0.3776
p=0.2655
p=0.0397
p=0.3208
p=0.6384
p=0.7977
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2